

Lessons Learned in Inclusive Game Design: Bridging the Gaps for Visually Diverse Players

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Abstract

This paper presents a comprehensive study on the design, development, and evaluation of a multi-sensory board game, named Ezam, to bridge the participation gap between visually impaired and abled players. Using a user-centered, participatory design approach, the project integrates tactile, auditory, and spatial cues within a service-design framework to ensure a holistic, equitable gaming experience. The design process was informed by a market analysis of existing accessible board games and co-design sessions with over 50 visually impaired and sighted stakeholders across educational and community settings in India, and refined through iterative prototyping and rigorous user testing. Quantitative and qualitative evaluations indicate significant improvements in usability and engagement, while pinpointing areas for further refinement. This study advances inclusive design by (1) positioning physical games as service experiences, (2) illustrating how multi-sensory mechanics foster equity among visually diverse players, and (3) offering participatory strategies and design guidelines for accessible, multi-sensory board games.

Keywords: Inclusive design, multi-sensory game design, visually impaired, service design for accessibility

1. Introduction

Physical board games continue to hold deep cultural and social importance by bringing people together through shared, hands-on play. Yet most are still built around visual elements, which creates barriers for players with visual impairments. In India, where the Rights of Persons with Disabilities Act (Government of India, 2016) guarantees equal participation in recreational and cultural activities, this exclusion is particularly pronounced. Despite advances in assistive technologies, most board games continue to rely on visual cues, inadvertently sidelining the nearly 285 million individuals globally affected by visual impairments (WHO: Newsroom, 2023). Truly inclusive board games remain rare, and when they do exist, they often treat accessibility as an afterthought, relegating visually impaired players to dependent roles rather than offering an equal experience (Yuan et al., 2011).

This paper looks at a different approach: designing a board game that both sighted and visually impaired players can enjoy competitively and independently, without either group gaining a sensory advantage. Our goal was to show that realizing this vision means thinking beyond just product features or basic accessibility. It demands a shift toward service design, which emphasises holistic, user-centered experiences unfolding across all moments of interaction (Stickdorn & Schneider, 2010).

Drawing on Stenros & Sotamaa's player service model (2009), which delineates five types of game-related services that support play and Huunonen's insights into the player journey (2015), we employed participatory, user-centered methods to co-create a multi-sensory game with both visually impaired and sighted participants. Huunonen argues that game designers can learn a great deal from service-design principles, especially when considering the player's end-to-end experience (Huunonen, 2015, pp. 8–11). This process highlighted how inclusive game design can serve as a testing ground for broader service-design approaches rooted in empathy, iteration, and equitable experience delivery. We deliberately designed entry points, ongoing interactions, social exchanges, and moments of reflection, using touch, sound, spatial layout, and shared rules, to ensure every player can engage meaningfully and fairly through a more balanced form of play (Stickdorn & Schneider, 2010).

By reframing inclusive game design as service design, we move beyond “fixing” accessibility toward reimagining the entire experience. This paper shares key lessons on tactile and auditory strategies that promote fairness, the social impacts of multi-sensory mechanics, and participatory approaches that foreground lived experience. In doing so, we contribute to three areas:

1. Repositioning inclusive physical game design within the broader practice of service design, and illustrating how the game design process itself offers lessons for service-design methodologies.
2. Demonstrating how multi-sensory mechanics can foster equitable experiences among visually diverse players.
3. Offering participatory strategies rooted in empathy, iteration, and equitable experience delivery.

2. Related work and literature review

This section critically examines prior research on inclusive game design, adaptations of accessible games, and participatory design methodologies. The review delineates the theoretical foundations and empirical findings that inform our study, with an emphasis on multi-sensory strategies and user-centered design approaches in both digital and physical gaming contexts.

2.1 Inclusive Game Design

Inclusive game design emphasises the integration of accessibility as a core principle rather than as an afterthought. Foundational work by Yuan et al. (2011) highlights that traditional games rely heavily on visual cues, which inadvertently exclude visually impaired players. In response, recent studies advocate for multi-sensory feedback (incorporating tactile, auditory, and spatial cues) to create more accessible experiences. For example, research on audio-tactile interactions in gaming demonstrates that integrating non-visual cues can significantly enhance usability for visually impaired users (Archambault & Olivier, 2005). Additionally, the Game Accessibility Guidelines (n.d.) offer a comprehensive framework for incorporating accessibility into game design. Building on these insights, our project explores the

application of multi-sensory design principles to a physical board game, a domain that remains largely underexplored.

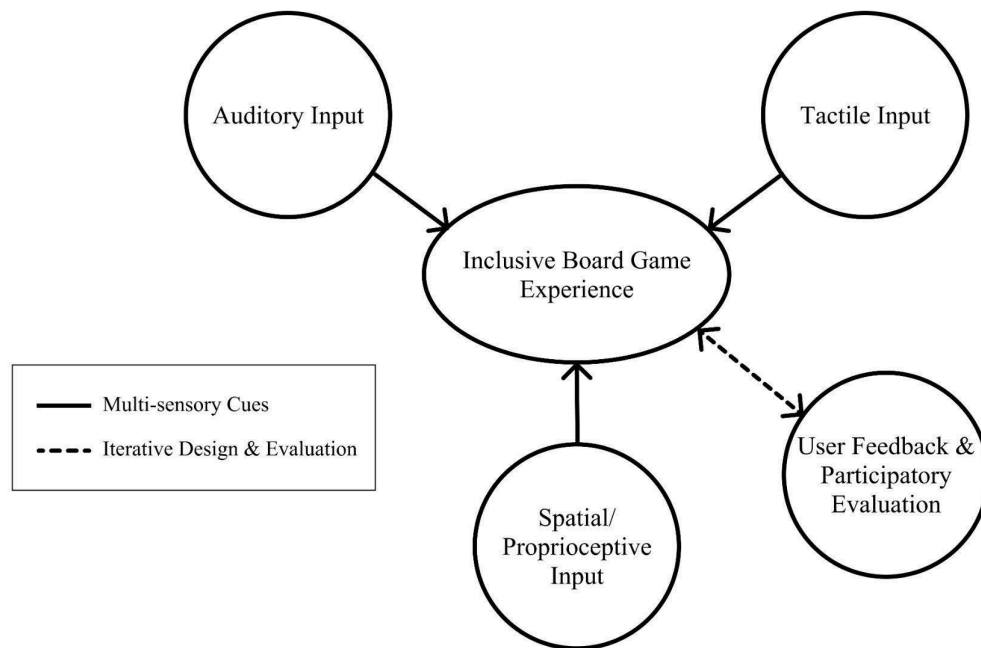


Figure 1. Conceptual Framework illustrating how tactile, auditory, and spatial inputs converge to form an inclusive board game experience, with an iterative feedback loop guided by user insights.

In the Indian context, Gupta et al. (2021) documented a participatory design process involving visually impaired children in Mumbai, emphasizing the value of direct user involvement in shaping tactile interfaces. Their findings highlight how co-designing with users led to better adoption of form elements and more intuitive tactile interaction. A cross-sectional study by Senjam et al. (2019) surveyed 250 students across 10 blind schools in Delhi and found that tactile-based assistive technologies (e.g. Braille slate, abacus) were widely available and used, whereas more advanced visual-based aids were scarce or absent, even among students with residual vision . Moreover, only 11 out of 52 evaluated technologies had coverage in at least 60% of schools, highlighting a clear gap in access to diverse AT devices. These findings informed our design to prioritise low-cost, tactile and auditory components proven to

be widely familiar and feasible, rather than relying on technologically complex features that may be unavailable in many Indian school settings.

2.2 Review of Accessible Games

The evolution of accessible games has seen numerous adaptations of classic board and video games to better accommodate users with disabilities. Research has focused on modifying games such as Ludo, Chess, and Othello to address inherent accessibility limitations. Valério Neto et al. (2020) detail an educational game adapted for visually impaired users, highlighting the importance of audio-centric design and tactile modifications. Furthermore, the study "How to Make Games for Visually Impaired Children" (Archambault & Olivier, 2005) discusses the adaptation of game mechanics by leveraging auditory and tactile feedback. These works collectively reveal that while digital adaptations have advanced accessibility, similar multi-sensory principles have not been sufficiently applied to physical board games. A notable Indian precedent is "Adu Huli", a tactile adaptation of a traditional board game recognized through the SRISTI Gandhian Young Technological Innovation (GYTI) awards (2014). The redesign incorporated tactile engravings, peg-based spatial cues, and texture variations to make the game playable without visual input, emphasizing affordability and inclusive play in low-resource schools. Our research builds upon this emerging direction by developing a tactile, multi-sensory board game aimed at fostering equitable participation between visually impaired and sighted players, while addressing gaps in mainstream physical game design.

2.3 Existing Accessible Board Games

In parallel with our review of the academic literature, we conducted a market analysis of popular board games that have been adapted for or marketed as accessible to visually impaired players. This examination revealed both the potential of physical board games as inclusive platforms and the persistent design shortcomings that impede full participation. Below, we summarize our analysis of four commonly referenced games—Ludo, Chess, Othello, and various products from Masters Traditional Games—to illustrate the limitations that informed our own design decisions. Furthermore, as part of our user research, we will collect user feedback

and experiences related to these games to gain a deeper understanding of their engagement factors and usability.

1. Ludo

Despite being simple in concept and widely recognized, Ludo poses several challenges for visually impaired users. The counting of steps, even though tactile, is significantly more difficult for visually impaired than for those who can count visually. Moreover, sighted participants are required to provide extra assistance, which can lead to an imbalanced experience and trigger empathy or sympathy factors rather than promoting equitable engagement. Although certain adaptations (such as tactile dice) mitigate issues like dice tumbling and getting lost, the core mechanics of Ludo still hinge on visual differentiation.



Figure 2. Ludo Board Game in Braille (by Advance Engineering Works, Product code: 20112)

2. Chess

Chess is frequently cited as a tactile-friendly board game, thanks to specialized boards and pieces with raised or recessed elements. However, our analysis indicated that these adaptations can remain cumbersome: boards are often too small for comfortable tactile navigation, and the precise fitting or repositioning of pieces is time-consuming. Sighted players may find the slowed pace less enjoyable, inadvertently reinforcing a sense of “special accommodation” rather than genuine inclusivity.

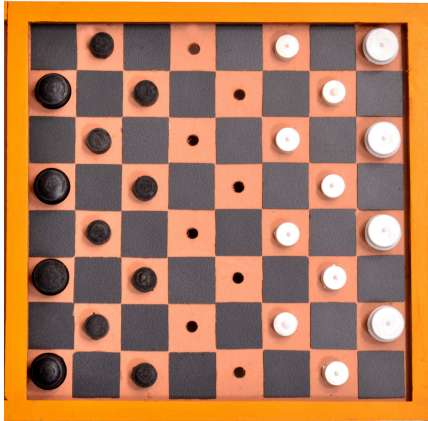


Figure 3. Wooden Simple Chess Board (by Advance Engineering Works Product code: 20107B)

3. Othello

Othello relies heavily on pattern recognition, an aspect that grants visually abled players a distinct advantage. Even with reduced-scale boards or contrasting textures on the pieces, visually impaired users still face difficulties tracking the evolving patterns. Paradoxically, scaling down the board to simplify tactile exploration may render the game less challenging for sighted players, who can quickly lose interest. Thus, while tactile versions exist, they have not fully resolved the balance between accessibility and sustained engagement.



Figure 4. Reversi Othello Board Game (by Crunchyroll Store)

4. Masters Traditional Games

A range of accessible board games is available through specialized retailers such as Masters Traditional Games. These products typically integrate raised grids or Braille-like markings, yet they remain prohibitively expensive and difficult to obtain, particularly in many parts of India. Moreover, despite the incorporation of tactile elements, most of these adaptations still depend significantly on color or visual cues. As a result, they can inadvertently perpetuate the need for external assistance or sighted oversight.

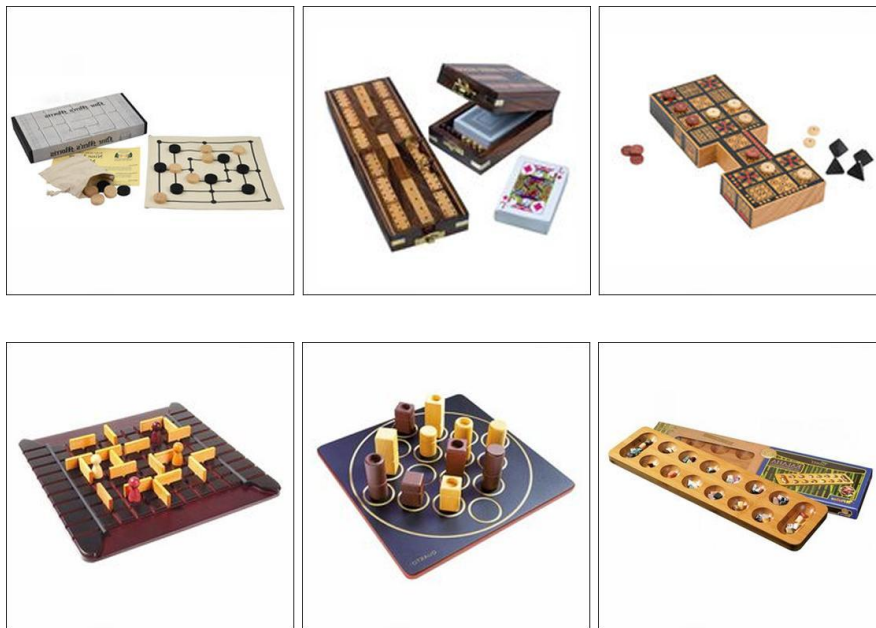


Figure 5. Range of accessible board games (by Masters Traditional Games)

Implications for Our Design

Taken together, these observations highlight the need for a more holistic approach to accessible board game design, beyond isolated tactile add-ons or color tweaks. Existing products often offer only single-sense enhancements, command excessive prices, and rely on sighted assistance, treating accessibility as an afterthought rather than embedding it in core mechanics. By pinpointing these gaps, we identified key areas—such as balanced tactile navigation, intuitive auditory feedback, and robust physical interfaces—that informed the multi-sensory strategies adopted in our own board game design.

2.4 Synthesis of Accessible Game Literature & Study

The literature and market analyses collectively illustrate the progress and limitations in accessible game design. While prior research underscores the effectiveness of multi-sensory strategies and participatory methodologies, particularly in digital contexts, our review of physical board games reveals that adaptations are often fragmented (lacking integrated tactile and auditory features), expensive or frequently depend on external facilitation (as seen in examples like Ludo and Chess).

By integrating these observations, we identify a critical gap: although board games have the potential to provide an inclusive, shared experience for visually impaired and abled players, the current market rarely implements cohesive, user-centered design solutions. Moreover, studies of digital adaptations and the insights from participatory design approaches suggest that collaborative, iterative processes are crucial for developing engaging and equitable gameplay.

3. Methodology

This section outlines the methodological framework adopted for our study, which is grounded in a user-centered, participatory design approach. The framework guided our iterative process and informed the collection of both qualitative and quantitative data, ensuring that our design effectively addressed the accessibility needs of visually impaired players while engaging abled users.

3.1 Research Approach and Framework

Our study was conducted in accordance with established principles of user-centered design (ISO, 2019) and participatory design. We engaged end users (visually impaired people, caregivers, teachers, and family) as active collaborators, incorporating their feedback at every stage of development. The design process was structured as an iterative cycle, comprising the stages of empathizing with users, defining core challenges, ideating potential solutions, prototyping designs, and validating these solutions through user testing. While these phases are well established in design research (Brown, 2009; Plattner et al., 2009), we adapted this framework to suit the specific challenges of designing a tactile, multi-sensory board game for visually impaired players in India.

3.2 Data Collection Methods

We employed a mixed-methods strategy to ensure a comprehensive understanding of user needs and system performance. A sample of more than 50 participants were involved in the interviews & focus group discussion.

1. Interviews

Individual interviews were conducted with visually impaired participants to capture their social experiences, accessibility challenges, and expectations. Additionally, expert interviews with educators, caregivers, trainers, family, and friends of the visually impaired provided contextual insights into the practical aspects of tactile and auditory design.

2. Focus Groups

Focus groups were organized with visually impaired players and their families to facilitate collaborative discussions. These sessions enabled participants to compare experiences and offer collective feedback on experience and iterations.

3. Behavioural Observations and User Testing

We conducted iterative testing sessions in naturalistic settings, recording participant interactions and behaviors while using the prototypes. Observation protocols and the think-aloud method (Courage & Baxter, 2005) were employed to capture detailed insights into user challenges and areas for improvement.



(a)

(b)

(c)

Figure 6. Interviews (b) and Focus Group Discussions (c) with visually impaired students and educators at Netraheen Kanya Vidyalaya, Jabalpur (a).

3.3 Participants and Stakeholders

Target Users

The primary participants were visually impaired teenagers and young adults (ages 13–21) from a blind school in Jabalpur (Madhya Pradesh), Bangalore (Karnataka), and friends/family in our network. These participants had varying degrees of visual impairment (from partial sight to total blindness) and diverse experiences with board games.

Key Stakeholders

Educators, Trainers and Caregivers: These stakeholders provided domain expertise on tactile literacy, spatial awareness, and the practical aspects of integrating multi-sensory feedback into game design. Their insights were crucial in establishing safe and effective testing environments.

Family Members and Friends: Family members and friends contributed practical perspectives on everyday challenges and the social dynamics of gameplay, highlighting how inclusive design can foster shared experiences between visually impaired and abled individuals, but only if incorporated thoughtfully.

By engaging a diverse group of participants and using an iterative design process, we developed a board game that meets the accessibility needs of visually impaired players while offering an engaging experience for all. In the following sub-section, we describe the participatory design philosophy that informed our collaboration with users, before moving on to prototype development and testing.

3.4 Participatory and User-Centered Design Methods

To complement the iterative design process described in Section 3.1, this subsection outlines the participatory design principles that underpinned our user engagement strategy. Involving end users directly in the design process is widely recognized as essential to ensuring that accessibility solutions meet real-world needs. Druin (2002) underscores the value of engaging children and other end users in co-design, arguing that such involvement is crucial for capturing authentic user requirements. Mattheiss et al. (2016) further illustrate the efficacy of user-centered design processes involving visually impaired pupils in yielding significant insights for the

development of accessible systems. Sanders et al. (2008) advocate for co-creation methods that empower diverse stakeholders by integrating their perspectives throughout the design cycle. Foundational human-computer interaction literature emphasise that iterative design cycles are critical for identifying and addressing accessibility issues (Lazar et al., 2017). In alignment with these approaches, we engaged visually impaired users, caregivers, and educators throughout multiple iterative prototyping and evaluation cycles, adapting co-design methods to the cultural and resource constraints of Indian settings.

3.5 Data Analysis

User Research Results

We conducted interviews and focus groups with over 50 stakeholders including visually impaired users, educators, caregivers, and family members. Thematic analysis uncovered insights to inform our inclusive game design.

- **Visually Impaired Users:** Participants expressed a strong preference for auditory and tactile games, emphasizing independence and fairness over assistance.
 - Auditory Confidence: Users favored riddles, music, and sound-based play. *“Ham sun ke sab pata kar lete hai.” (We can find out everything by listening.)*
 - Limited Outdoor Access: Many feared injuries due to unsafe environments. *“Khelne waqt toh idhar se udhar girte padte rehte hain.” (While playing, we keep falling all over.)*
 - Feeling Excluded: Social interactions with sighted peers often led to isolation. *“Main dosto ke sath jaata tha, toh mujhe side mein bitha dete the.” (When I went with friends, they would make me sit aside.)*
 - Desire for Independence: Users resisted unnecessary assistance. *“Hum kisise kam nahi hai.” (We are not less than anyone.)*
- **Stakeholder Perspectives**
 - Educators stressed the need for structured, voice-based games. *“Jo bhi banayenge, usme voice dalna hi padega.” (Whatever you make, you’ll have to include voice (feedback) in it.)*

- Caregivers and family highlighted emotional and social barriers, including the lack of peer engagement.

“Sighted kids interested nahi the kisi ko assist karne mein.” (Sighted kids were not interested in assisting anyone.)

- **Key Learnings**

- Accessibility must be paired with equity.
- Sound and touch are empowering mediums.
- Inclusive design reshapes group dynamics.
- Solutions must be simple, safe, and context-appropriate.

User Feedback on Existing Accessible Games

To complement our market analysis, we conducted an user evaluation of four commonly cited board games- Ludo, Chess, Othello, and selected titles from Masters Traditional Games (Section 2.3 Existing Accessible Board Games) through structured interviews with 35 visually impaired participants.

As physical copies were unavailable for hands-on testing, we provided detailed verbal descriptions of each game’s rules, tactile features, and common accessibility adaptations. Participants then reported prior experience and shared their impressions, focusing on three primary questions:

1. Have you played this game before, and if so, how frequently?
2. What challenges did you face while playing this game, if any?
3. Did you feel equally engaged compared to sighted players?

The results revealed that:

- 29 participants had played Ludo at least once, but 22 reported difficulty in tracking movement without assistance.
- 21 participants had experience with Chess, with 16 stating they enjoyed it due to its tactile-friendly nature, although 11 found the slow pace frustrating.
- Only 6 participants had played Othello, and most found it confusing or difficult to follow due to the reliance on patterns.

- 4 participants reported familiarity with products from Masters Traditional Games, typically through institutions or workshops; however, all six noted the lack of personal ownership due to high cost and limited availability.

These insights confirmed that existing adaptations often fail to deliver a fully independent and engaging experience. They directly informed our design priorities- tactile clarity, intuitive layout, and equitable engagement across abilities.

4. Design process and prototype development

This section details our iterative design process for developing a multi-sensory board game, from initial ideation informed by user insights and market research to the final prototype.

4.1 Conceptualization and Ideation

Our initial design phase was driven by insights gathered from interviews, focus groups, and market analysis of existing accessible board games (as detailed in Section 2.3). Insights from visually impaired users, educators, and families highlighted the need for tactile clarity, auditory cues, and non-reliance on visual elements. These insights were synthesized, brainstorming sessions were held and we generated multiple design concepts.

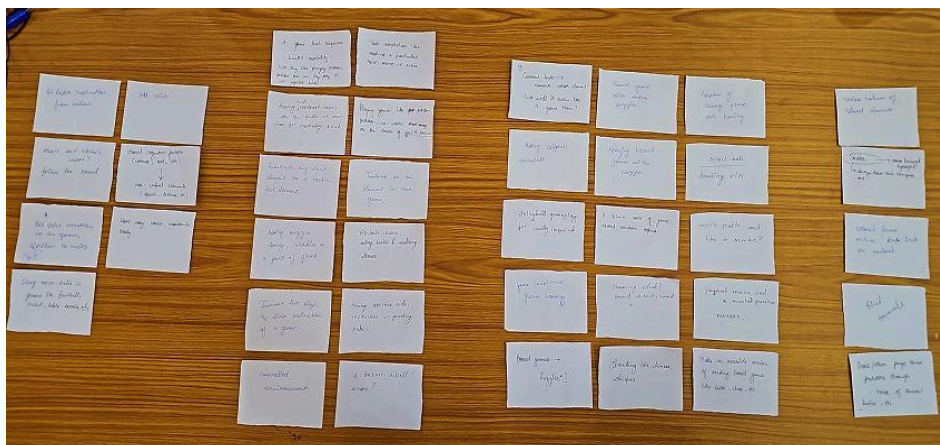


Figure 7. Rapid brainstorming session (approximately 40 ideas generated)

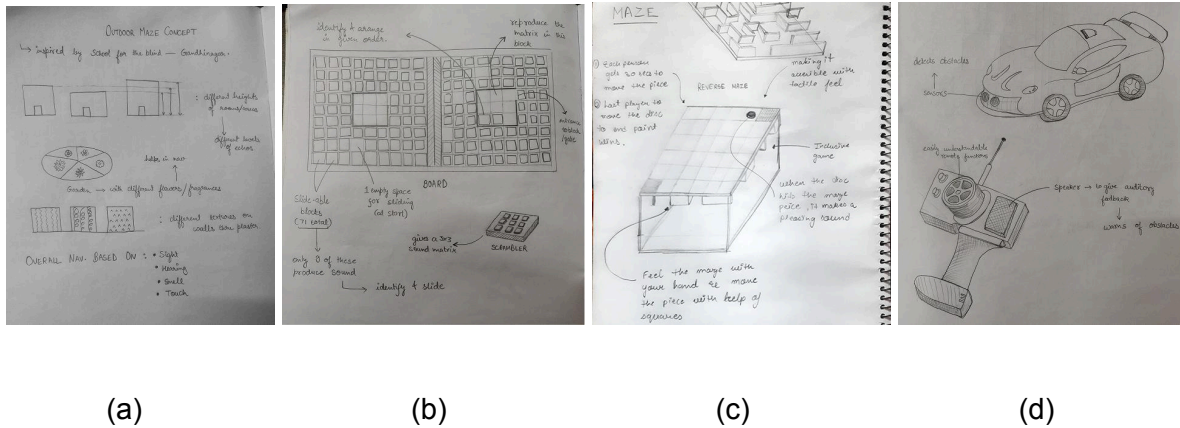


Figure 8. First set of ideations/concepts (a) An outdoor maze concept; (b) Sound based puzzle; (c) Board-based maze concept; and (d) Voice operated car.

Ultimately, our ideation process converged on a design that integrated elements from four distinct initial concepts, resulting in a board game built around a chase mechanism within an inverted maze (Ezam). Outdoor formats were excluded due to unresolved safety concerns for visually impaired participants, and we opted not to incorporate electronic sound components to keep manufacturing costs manageable.

4.2 Prototype Development and Iterative Refinement

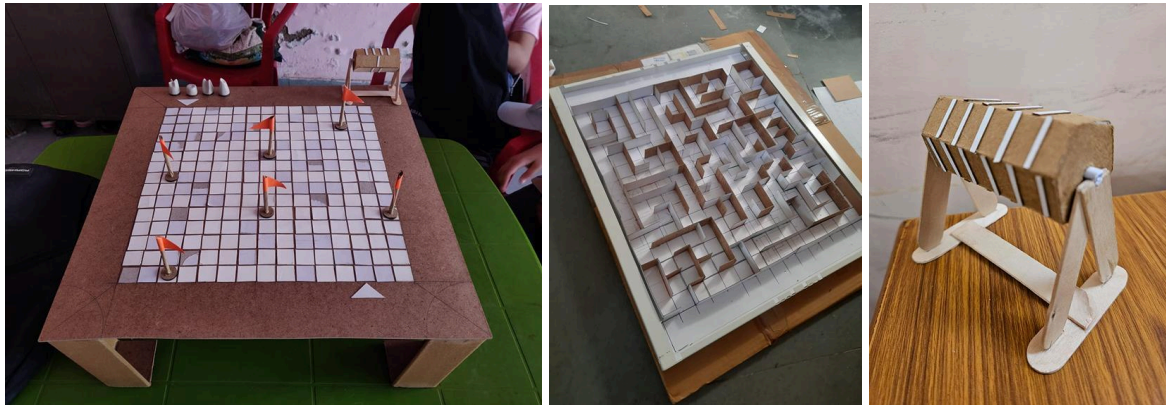
We developed low-fidelity prototypes using materials like paper and foam to quickly test tactile navigation. Early feedback revealed issues with indistinct board textures and die readability. Based on user input, we iteratively refined the design using durable materials and clearer tactile elements, ensuring each version improved accessibility and balanced gameplay for both visually impaired and sighted users.

4.3 Final Prototype and Implementation Details

The final prototype represents a culmination of iterative refinements that integrated tactile, auditory, and spatial design elements into a cohesive board game. The design features a tactile board with raised grids and Braille markings, complemented by embedded auditory cues that signal game events (e.g., dice rolls and player movements). Spatial elements are addressed through the careful arrangement of game components, ensuring that the board layout facilitates easy navigation for visually impaired users without compromising the engagement of sighted players.

Technical specifications of final product include:

1. Materials: Durable wood for the pieces, die and board/table (with engraved tactile markers), and mini bells for auditory feedback.
2. Design Features:
 - Raised grids and walls to support tactile exploration.
 - Embedded bells to provide immediate auditory feedback when game pieces/die is moved.
 - A modular design allowing easy reconfiguration of game components (flags and pieces) based on user preference



(a) Board Top

(b) Board bottom (maze)

(c) Die

Figure 9 (a, b & c). Final Prototype made from hardboard and cardboard



(a) Top

(b) Bottom

Figure 10 (a & b). Digital visualisation of final product (Ezam)

5. Evaluation and results

This section details the evaluation procedures and the results from testing Ezam. Both quantitative and qualitative data were collected to assess usability, user engagement, and overall accessibility for visually impaired and sighted players.

5.1 User Testing Setup and Data Collection

User testing was conducted in controlled, naturalistic settings at the educational institutions in Jabalpur and Bangalore. Participants included visually impaired teenagers and young adults, along with educators, caregivers, and family members. The testing environment was arranged to mimic typical gameplay (school, home, etc.) while ensuring participant safety and comfort.

Participants were assigned tasks such as navigating the board, interpreting tactile cues, and responding to auditory signals. Data were collected using the following methods:

1. **Interviews:** Semi-structured individual interviews were conducted after testing sessions to capture detailed feedback regarding usability and overall game experience.
2. **Behavioural Observations:** We employed think-aloud protocols (Courage & Baxter, 2005) and observation checklists to record user interactions, errors, and adaptive behaviors during gameplay.



(a) Feeling the game

(b) Understanding Gameplay

(c) Playing the game

Figure 11. Testing with the users (at Netraheen Kanya Vidyalaya, Jabalpur)

5.2 Results

Quantitative Results

A total of 22 (19 visually impaired + 3 visually abled) participants completed both early and final prototype testing sessions.

- The mean satisfaction rating was 4.2 out of 5, with no participant rating below 2.8.
- Time taken by participants to complete one full round of gameplay decreased from an average of 42.4 minutes in the initial version to 29.6 minutes in the final prototype- an improvement of about 30%.
- Tactile cue interpretation (counting & navigating) errors dropped from an average of 3.1 per session to 1.2 after refinement of raised markings.
- Error rate reductions were consistent across both fully blind and low-vision participants.

Qualitative Results

Thematic coding of interviews and focus group transcripts revealed several recurring observations:

- 19 of 22 participants reported that tactile and auditory cues supported greater independence during gameplay.
- 17 of 19 visually impaired users valued the raised grid layout, noting improved spatial awareness.
- 3 of 3 sighted players described the multi-sensory elements as engaging and fair, without adding unnecessary complexity.
- 11 of 22 participants suggested improvements in auditory cue clarity and spatial direction, especially during group play.

These responses informed subsequent design iterations and are later expanded upon in the discussion section.

6. Discussion

This section synthesizes the insights derived from our design and evaluation phases, discusses their broader implications for inclusive board game design, and outlines the study's limitations along with future research directions.

6.1 Interpretation of Key Findings

The evaluation yielded several specific insights that reflect both the successes and remaining challenges of our inclusive board game design:

1. Enhanced Navigation and Spatial Awareness

- Smooth Tactile Guidance: Participants with visual impairments reported being able to navigate the inverted maze layout with minimal assistance once they became familiar with the tactile markers.
- Room for Step-Counting Feedback: Multiple users specifically requested a mechanism, either auditory or tactile, to confirm how many steps each pawn had moved, underscoring the need for clearer positional feedback or movement tracking.

2. Improved Usability

- Reduced Completion Times: Measurable improvements were noted after adjustments to board spacing and tactile indicators, with user testing sessions. These improvements reflected users' increasing confidence and spatial memory after refinements.
- Decreased Confusion Rates: Observational data indicated a roughly 24% decline in user confusion related to locating ponds or verifying pawn positions, though participants still highlighted the challenge of tracking movement when pawns were grouped closely.

3. Balanced Engagement for Both User Groups

- Inclusive Competition: Sighted players appreciated that the game remained competitive and did not feel artificially slowed down by accessibility features. This synergy was particularly evident in group play, where visually impaired and sighted participants reported positive social dynamics.

- Sustaining Interest: While many sighted users found the tactile elements intriguing, some recommended further refinements (e.g., streamlined board space, more direct feedback on movement) to maintain long-term engagement.

4. Concrete Design Refinements

- Board Layout and Pawn Dimensions: Users suggested reducing empty space on the board to eliminate dead zones and making pawns taller and thinner for easier handling.
- Ongoing Ownership and Accessibility: Several visually impaired users inquired about retaining the game post-study, underscoring both its perceived value and the broader need for affordable, permanently accessible board games.

Together, these findings affirm the effectiveness of a multi-sensory design approach and provide actionable directions for refinement.

6.2 Relevance to Service Design

The design, implementation, and evaluation of the Ezam board game offer direct, actionable insights for inclusive service design—not just in gaming—but in broader contexts such as education, public services, and shared recreational activities. The process maps closely onto core service design principles, as discussed by Stenros & Sotamaa (2009), in the following ways:

- Maintenance of Environment: Arrangement of tactile elements and the durability of materials ensured the board's usability for repeated, safe interactions. Similarly, accessible service environments require consistent upkeep to remain truly inclusive for all users.
- Support of Initiation: Tactile orientation points helped users immediately understand game setup and rules, paralleling onboarding strategies in inclusive services that ensure all users can begin participating independently and confidently.
- Facilitation and Assistance of Play: Integrated, real-time feedback (raised grids) allowed users to correct themselves and stay engaged without external help. Such embedded assistance is essential in service experiences, enabling users to navigate and interact autonomously.

- **Socialization of Players:** The game's multi-sensory and balanced competitive design fostered positive group dynamics between visually impaired and sighted participants. In service design, facilitating equitable social participation strengthens community and enhances user satisfaction.

In summary, the iterative and participatory co-design approach underlying the Ezam prototype exemplifies how service design can move from generic accessibility compliance to delivering genuinely engaging, independent, and social experiences for diverse user groups. The principles proven effective in this game inform best practices for accessible, user-centered service delivery in a wide range of contexts.

6.3 Implications for Inclusive Design

This study contributes to the theoretical discourse and practical application of inclusive design in multiple ways. First, it demonstrates that multi-sensory strategies, long-applied in digital accessibility (Yuan et al., 2011), can be effectively translated to physical board games, enabling equitable gameplay for visually impaired and sighted users alike. The positive reception of tactile markers and auditory feedback suggests that such strategies support independent navigation and fair competition without compromising the game's challenge or engagement.

Second, our findings reinforce the value of iterative, participatory design methods, particularly in resource-constrained contexts like India. By involving users early and consistently (across prototyping, testing, and refinement) we surfaced practical insights that might otherwise have been overlooked by expert-led evaluations alone. This aligns with co-design principles advocated by Sanders et al. (2008) and the user-centered guidelines of Gulliksen et al. (2003), but extends them to the development of low-tech, analog formats.

Finally, the work emphasises a context-sensitive approach to accessibility, one that balances innovation with feasibility. Design decisions were informed not only by usability but also by affordability, availability, and user familiarity—factors that are often underexplored in mainstream inclusive design literature. By grounding our approach in both empirical feedback and prior Indian AT research, we offer a replicable and theory-informed framework for inclusive product development in under-resourced settings.

6.4 Limitations and Future Directions

While our study presents promising results, several limitations warrant discussion. The participant sample was geographically limited to select urban centers in India, potentially affecting the generalizability of our findings across diverse cultural and socio-economic contexts. Future studies should aim to include a broader and more varied demographic to account for regional differences in accessibility, education, and exposure to assistive tools.

Additionally, although the (tactile and auditory) feedback mechanisms were effective, real-time movement feedback, such as step-count confirmation, remained an area of user concern. While we intentionally avoided electronics to maintain affordability, future iterations could explore low-cost hybrid or digital enhancements, (e.g., step counters or voice prompts) to improve real-time feedback without significantly increasing production costs.

Another limitation is the short-term nature of engagement. Since our evaluation took place during time-bound sessions, longitudinal studies could offer deeper insights into how inclusive board games like Ezam affect sustained engagement, social interaction, and educational outcomes. Field-testing in schools, community centers, or recreational programs may also help assess real-world adoption and impact.

These limitations also highlight future directions for research and development: (1) expanding demographic and contextual diversity, (2) refining multi-sensory feedback through cost-effective enhancements, and (3) evaluating long-term usability and integration of inclusive games in everyday contexts. Such efforts can further validate inclusive design as a scalable, context-responsive approach in low-resource environments.

7. Conclusion

This section summarizes the key contributions of the project, outlines potential directions for future work, and reflects on the broader impact of inclusive board game design.

7.1 Summary of Contributions

This study presents a comprehensive account of designing an inclusive board game that integrates tactile, auditory, and spatial elements to accommodate visually impaired players while engaging sighted participants. By employing a user-centered, participatory design approach, our work demonstrates that multi-sensory strategies, traditionally applied in digital game accessibility (Yuan et al., 2011), can be successfully translated to physical board games. Our iterative process, guided by direct feedback from visually impaired users, educators, and family members, resulted in a refined prototype that improved independent navigation, reduced confusion, and enhanced gameplay satisfaction across user groups. These results, supported by both qualitative feedback and quantifiable improvements (e.g., ~30% reduction in game completion time and decreased error rates), demonstrate that a balanced multi-sensory experience is critical to fostering equitable participation. These validated insights not only inform inclusive game design but also point toward future opportunities for scaling such approaches in diverse, resource-constrained contexts. Additionally, the process contributes actionable insights to inclusive service design, especially in contexts where equitable access, onboarding, and independent participation are central to user experience.

7.2 Final Remarks

This work advances the field of inclusive game design by demonstrating that physical, multi-sensory board games can bridge the gap between visually impaired and abled players. Our findings reinforce that participatory design, when grounded in real-world user contexts and supported by iterative feedback, can lead to measurable improvements in accessibility, usability, and shared user experience.

More broadly, our study contributes to inclusive design theory by extending Yuan et al.'s (2011) multi-sensory accessibility framework and Norman's (2013) principles of tangible affordances, both typically applied in digital domains, into low-cost physical games. It also advances co-design frameworks (Sanders & Stappers, 2008) by showing how these methods can be adapted to contextual challenges such as limited technology access, safety constraints, and varying levels of tactile literacy. By drawing parallels between game mechanics and service-delivery models, this work demonstrates how inclusive game design serves as a low-cost testbed for service design strategies that promote independent and equitable engagement.

As inclusive design continues to evolve, our study offers a replicable model for future projects seeking to create accessible and equitable interactive experiences. Together, these contributions advance both theory and practice, deepening our understanding of how tactile and auditory design can equalize play, while offering a replicable framework for practitioners designing for inclusion under resource constraints.

References

1. Blindness and vision impairment. (2023, August 10). Retrieved from <https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment>
2. Yuan, J., Folmer, E., & Harris, F. (2011). Game Accessibility: A Survey. *Universal Access in the Information Society*, 10(1), 81-100. <https://doi.org/10.1007/s10209-010-0189-5>
3. Mattheiss, E., Regal, G., Sellitsch, D., & Tscheligi, M. (2016). User-Centred Design with Visually Impaired Pupils: A Case Study of a Game Editor for Orientation and Mobility Training. *International Journal of Child-Computer Interaction*, 11, 12-18. <https://doi.org/10.1016/j.ijcci.2016.11.001>
4. Neto, L.V., Fontoura Junior, P. H. F., Bordini, R. A., Otsuka, J. L., & Beder, D. M. (2020). Design and implementation of an educational game considering issues for visually impaired people inclusion. *Smart Learn. Environ.* 7, 4. <https://doi.org/10.1186/s40561-019-0103-4>
5. Archambault, D., & Olivier, D. (2005). *How to make games for visually impaired children*. Paper presented at the Association for Computing Machinery, New York, NY, USA. Retrieved from <https://doi.org/10.1145/1178477.1178578>
6. Metatla, O., Bardot, S., Cullen, C., Serrano, M., & Jouffrais, C. (2020). *Robots for Inclusive Play: Co-designing an Educational Game With Visually Impaired and sighted Children*. Paper presented at the Association for Computing Machinery, New York, NY, USA. Retrieved from <https://doi.org/10.1145/3313831.3376270>

7. Druin, A. (2002). The role of children in the design of new technology. *Behaviour & Information Technology*, 21(1), 1-25.
<https://doi.org/10.1080/01449290110108659>
8. Game Accessibility Guidelines. (n.d.). Retrieved from
<https://gameaccessibilityguidelines.com>
9. Sanders, E. B. N., & Stappers, P. J. (2008). Co-creation and the new landscapes of design. *CoDesign*, 4(1), 5-18.
<https://doi.org/10.1080/15710880701875068>
10. Lazar, J., Feng, J., & Hochheiser, H. (2017). *Research Methods in Human-Computer Interaction (2nd ed.)*. Cambridge, MA, United States: Morgan Kaufmann (an imprint of Elsevier).
11. International Organisation for Standardisation. (2019). *Ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems* (ISO Standard No. 9241-210:2019).
<https://www.iso.org/standard/77520.html>
12. Brown, T. (2009). *Change by Design*. New York, NY, United States: HarperBusiness.
13. Plattner, H., Meinel, C., & Weinberg, U. (2009). *Design Thinking*. Munich, Germany: mi-Wirtschaftsbuch.
14. Courage, C., & Baxter, K. (2005). *Understanding Your Users: A Practical Guide to User Requirements Methods, Tools, and Techniques*. San Francisco, CA, United States: Morgan Kaufmann (an imprint of Elsevier).
15. Norman, D. A. (2013). *The design of everyday things: Revised and expanded edition*. New York, NY, United States: Basic Books. (UK edition published by MIT Press).
16. Gulliksen, J., Göransson, B., Boivie, I., Blomkvist, S., Persson, J., & Cajander, Å. (2003). Key principles for user-centred systems design. *Behaviour & Information Technology*, 22(6), 397-409.
<https://doi.org/10.1080/01449290310001624329>

17. Gupta, A., Fulfagar, L., & Upadhyay, P. (2021). Co-designing with visually impaired children. In A. Chakrabarti, R. Poovaiah, P. Bokil, & V. Kant (Eds.), *Design for Tomorrow—Volume 2* (Vol. 222, Smart Innovation, Systems and Technologies). Singapore: Springer.
https://doi.org/10.1007/978-981-16-0119-4_35
18. Senjam, S. S., Foster, A., Bascaran, C., Vashist, P., & Gupta, V. (2019). Assistive technology for students with visual disability in schools for the blind in Delhi. *Disability and Rehabilitation: Assistive Technology*, 15(6), 663–669.
<https://doi.org/10.1080/17483107.2019.1604829>
19. Surender, E. (2014). *Adu Huli: A game for visually challenged children* [Design project]. Gandhian Young Technological Innovation (GYTI) Awards, SRISTI. Retrieved from
<https://gyti.techpedia.in/project-detail/adu-huli-a-game-for-visually-challenged-children/2040>
20. Huunonen, L. (2015). Designing game services [Master's thesis, Aalto University]. Aalto University.
21. Stickdorn, M., & Schneider, J. (2010). *This is Service Design Thinking*. Amsterdam: BIS Publishers.
22. Stenros, J., & Sotamaa, O. (2009). *Commoditization of helping players play: The rise of the service paradigm*. Paper presented at the DiGRA 2009 Conference.
<https://doi.org/10.26503/dl.v2009i1.413>